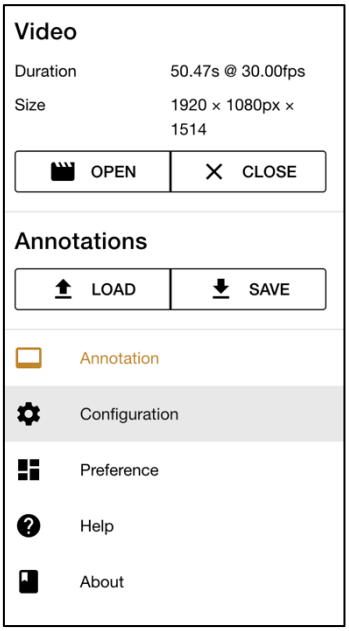
Annotation Guideline: MVRHOI



0. Preparation in Vidat:

- Upload a normalised video (.mp4)
- If applicable, upload the annotation status (.json)
- Upload the configuration file (.json)
- Ensure the frame rate is correct (30 fps)
- Set keyframes to 60 fps1.

1. Basic principle

Actions are described using interaction triples:

Subject → Action → Object

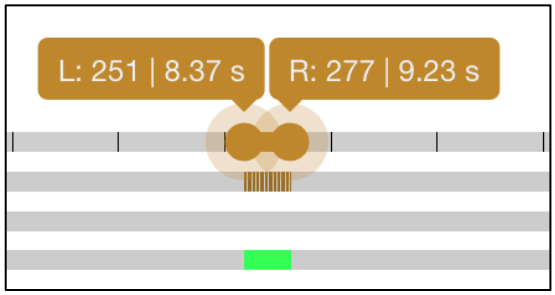
Annotate interactions in short sequences (2–5 seconds or approx. 90–160 frames).

In addition to the action annotations, spatial relations are also annotated to capture the context and dynamics of the interaction.

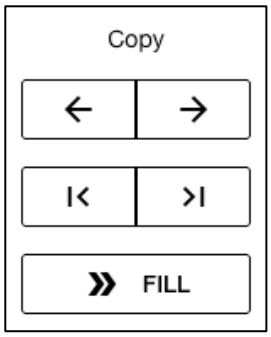
Select interactions that are as visually unambiguous as possible, with few irrelevant subjects or objects.

2. Procedure

Identify the interaction in the video (see the 'HOI Candidate List' sub-section for guidance). Define the start and end of the interaction.

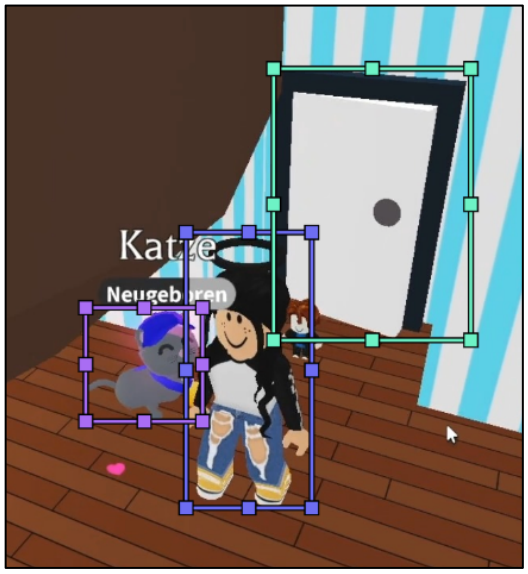


It is best to use L-frames and define subjects and objects in the start and end frames respectively (see Bounding Boxes), set instance IDs (see Subject Annotations), transfer them to the end frame using the Copy button and, if necessary, fill the gap using the Fill button (this automatically places markers in the intermediate frames, but is more suitable for static scenes and only works if all instance IDs have been set).



3. Bounding Boxes

Bounding Boxes: Create bounding boxes (square) for subjects and objects in every frame of the specified interaction period.



- Place the box as tightly as possible around the visible subject/object, without any parts of the subject/object extending beyond the box.
- For menus / UI elements: Only the relevant element, not the entire screen.
- For hands: Annotate each hand separately.

3. Subject-Annotations



Instance IDs (integers, numerical) are assigned consecutively for each video. The same character has a constant instance ID. Instance IDs must be unique within each video. The instance ID ‘0’ is reserved for the first-person perspective in every video.

In the event that the subject is not visible (e.g. during menu interactions), the following procedure applies:

Situation	Perspective	Subject	Instance	Create Bounding Box
Avatar or Person visible	3rd	„Person“	Dynamic	Yes
Person not visible	3rd	„User“	0	No
Egoperspective, hands visible	1st	„Hand“ (left/right)	Dynamic	Yes, per hand
Egoperspective, hands not visible	1st	„User“	0	No

4. Actions and relations

- Record the primary action (e.g. person-press-button).
- Annotate the spatial context before, during and after the action.
- Allow for multiple relations: spatial and action relations may occur simultaneously.
- Ensure frame accuracy: each relation has defined start and end frames.



4.1 Interactions

Examples:

- Person → holds → weapon
- Hand → taps → menu
- First-person user → opens → menu

List of possible interactions (selection):

open (1), close (2), press (3), turn (4), pull (5), pick_up (6), drop (7), carry (8), insert (9), use (10), ride (11), drive (12), climb (13), throw (14), shoot (15), collect (16), sit_or_lie (17), build (19), place (20), craft (21), write_on (22), erase_on (23), hit (24), kill (25), feed (26), greet (27), reload (28), open_menu (29), close_menu (30), enter_text (31), click_menu (32), look_at (41), watch (42), touch (43), hold (44), release (45)

4.2 Interactions with spatial relations

Annotation of positioning:
above (33), next_to (34), behind (35), away (36), towards (37), in_front_of (38), inside (39), beneath (40), on (46), under (47)

4.3 Duration: Interaction begins with the subject’s focus / attentiveness

4.4 Description: First, enter the subject's instance ID here, followed by the separator ';' (semicolon), followed by the object's instance ID.

5. HOI Candidate List:

The annotation focuses on the following core interaction classes:

1. **open→door**
2. **open→container** (box/chest/drawer/refrigerator/oven/window/shaft)
3. **close→container** (drawer/door/window/...)
4. **press→button_switch** (pushes button / flips switch / clicks hardware-switch)
5. **turn→knob_valve**
6. **pull→lever_handle** (inkl. antenna, handles)
7. **pick_up→hand_object** (grabs/takes/picks_up: cup, bottle, dice, plate, tape, book, tools, cards, coin, key, axe, health_pack, orb, rifle, ammo, tank,card,snowball,power_cell ...)
8. **drop→hand_object**
9. **carry→portable** (box, sword, bow, shield, bulb, bag ...)
10. **insert→keycard** (key + card)
11. **use→tool_device** (terminal, vending_machine, workbench, water_dispenser, keyboard, apparatus, mailbox, arcade, elevator)
12. **ride→rideable** (ride/jetski/boat/hot_air_balloon/mount/glider)
13. **drive→vehicle** (car/bus/motorcycle/tractor)
14. **climb→structure** (ladder/stairs/slide)
15. **throw→projectile** (dice/tomato/orb/disc/fishing_rod cast)
16. **shoot→weapon** (rifle/bow; inkl. shoots→lock)
17. **sit_or_lie→furniture** (sit_on chair/bench/couch/cushion; lie_in bed)
18. **build/place/craft→structure_item** (builds wall/places block/crafts item)
19. **write_on→surface** (draws_on window / sprays wall)
20. **erase_on→surface** (erases_from window)
21. **hit→character** / **kill→non_human**

UI/HUD

1. **open→menu** (incl. inventory)
2. **close→menu** (incl. inventory)
3. **enter→text**
4. **click→menu**

6. Example-Workflow with spatial relations

Climbing Sequence:

- Frames 5655–5658: person next_to structure (approaching)
- Frames 5659–5665: person climb structure + person on structure
- Frames 5666–5670: person on structure

Button-Press Sequence:

- Before Action (t-1): person next_to button, person look_at button
- During Action (t): person press button, person touch button, person in_front_of button
- After Action (t+1): person release button, person next_to button